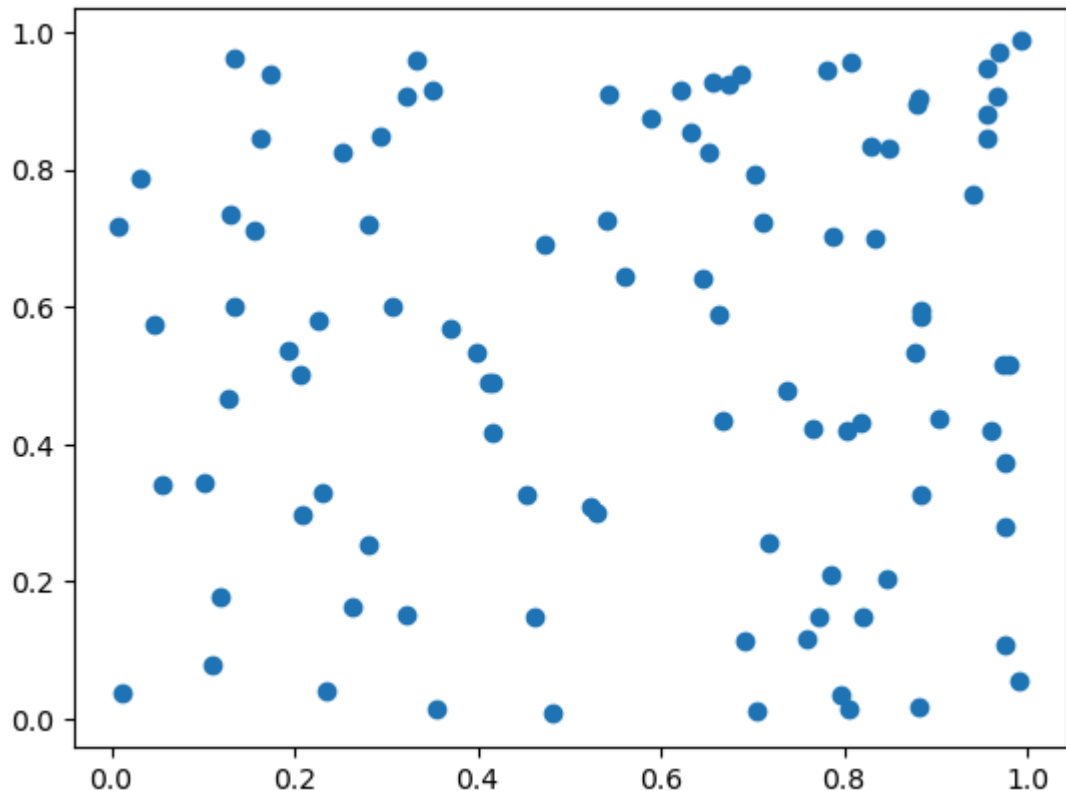


```
In [5]: import numpy as np
        from sklearn.cluster import KMeans
        import matplotlib.pyplot as plt
```

```
In [6]: X=np.random.rand(100,2)
```

```
In [7]: plt.scatter(X[:,0], X[:,1])
        plt.show()
```



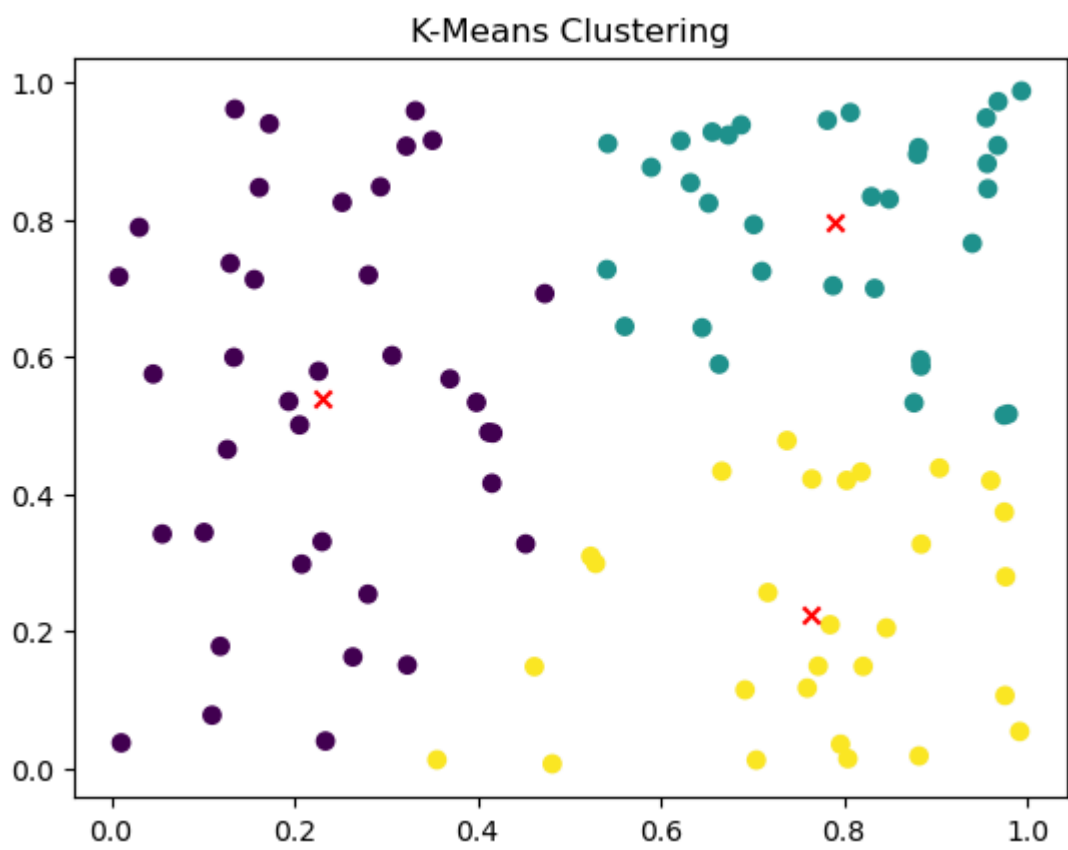
```
In [9]: kmeans=KMeans(n_clusters=3, random_state=0)
        kmeans.fit(X)
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(

```
Out[9]: ▼ KMeans ⓘ ?
        ► Parameters
```

```
In [10]: centroids=kmeans.cluster_centers_
         labels = kmeans.labels_
```

```
In [11]: plt.scatter(X[:, 0], X[:,1], c=labels)
         plt.scatter(centroids[:, 0], centroids[:, 1], marker='x', color='red')
         plt.title("K-Means Clustering")
         plt.show()
```



In []: